

Renewed Architecture: Verified Connectors + AI Orchestration + Governance-First UI

Layer 1: Verified Connector Library

What we build:

connectors/

└─ core/

| └─ base_connector.py # Standard interface all connectors implement

| └─ connector_registry.py # Discovery and management

| └─ auth_handler.py # OAuth, API keys, service accounts

└─ tier1/ # Build first (most requested)

| └─ stripe/ # Payments - transactions, refunds, disputes

| └─ slack/ # Notifications - channels, DMs, alerts

| └─ email_smtp/ # Email - send, templates

| └─ aws_s3/ # Storage - read, write, list

| └─ jira/ # Ticketing - create, update, query

| └─ salesforce/ # CRM - contacts, opportunities, cases

└─ tier2/ # Build second

| └─ paypal/

| └─ quickbooks/

| └─ hubspot/

| └─ zendesk/

| └─ twilio/

└─ tier3/ # Industry-specific

└─ epic_fhir/ # Healthcare EHR

└─ bloomberg/ # Financial data

└─ servicenow/ # IT operations

Each connector provides:

- Standardized actions (read, write, query, subscribe)

- Schema definitions (what data it handles)
 - Rate limiting built-in
 - Retry logic
 - Audit logging (every call recorded)
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Layer 2: AI Orchestrator

What we build:

orchestrator/

```
├─ intent_parser.py      # Understands "detect fraud in Stripe"
├─ workflow_designer.py  # Converts intent to workflow DAG
├─ connector_selector.py # Picks right connectors for the job
├─ rule_generator.py     # Creates business rules from natural language
└─ deployment_manager.py # Handles testing → staging → production
```

How it works:

User: "Alert me on Slack when Stripe charges over \$10,000 fail"

AI Orchestrator:

1. Intent: Monitor + Alert pattern
 2. Source: Stripe connector (charge.failed webhook)
 3. Filter: amount > 10000
 4. Action: Slack connector (send message)
 5. Generates workflow YAML + deploys
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Layer 3: Governance-First UI (The Differentiator)

Current state: Governance exists in backend but is invisible to users.

New state: Every screen shows governance in action.

New UI Structure:

Dashboard (index.html)

- └─ System Health
- └─ Active Agents with Risk Scores
- └─ Policy Violations (last 24h)
- └─ Pending Approvals queue
- └─ Compliance Score by vertical

Workflow Builder (builder.html) - REDESIGNED

- └─ Step 1: What do you want to do? (natural language)
- └─ Step 2: AI suggests workflow + connectors
- └─ Step 3: Configure with VISIBLE governance
 - └─ Risk tier badge on every action
 - └─ Required approvals highlighted
 - └─ Data handling policies shown
 - └─ Compliance tags (SOX, HIPAA, etc.)
- └─ Step 4: Sandbox Test (with simulated data)
- └─ Step 5: Review Evidence Trail → Deploy

Governance Center (NEW - governance.html)

- └─ Active Policies (what rules are enforced)
- └─ Risk Dashboard (real-time scores)
- └─ Approval Queue (human-in-the-loop)
- └─ Evidence Vault (searchable audit trail)
- └─ Compliance Reports (one-click export)

My Agents (agents.html) - ENHANCED

- └─ Each agent shows:
 - └─ Status + Risk Score

- | |— Policy compliance %
 - | |— Actions taken (with evidence links)
 - | |— Blocked actions (and why)
 - | |— Approval history
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Build Sequence

Phase 1: Foundation (Connectors + Governance UI)

1. Build connector SDK with standard interface
2. Build 3 Tier-1 connectors: Stripe, Slack, Email
3. Build Governance Center UI (governance.html)
4. Enhance Dashboard with governance widgets

Phase 2: AI Orchestrator

5. Build intent parser (natural language → structured intent)
6. Build workflow designer (intent → DAG)
7. Redesign builder.html as conversational wizard
8. Build sandbox test environment

Phase 3: Complete Connector Library

9. Build remaining Tier-1 connectors
10. Build Tier-2 connectors
11. Add connector marketplace UI

Phase 4: Polish

12. One-click compliance reports
 13. Approval mobile notifications
 14. Evidence export for auditors
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User Journey (End State)

1. User lands on Dashboard
 - Sees "Create Workflow" button prominently
2. Clicks "Create Workflow"
 - Natural language input: "Monitor Stripe for failed charges"

over \$5000 and alert finance team on Slack"

3. AI Orchestrator responds:

"I'll set up a workflow that:

- Monitors Stripe for charge.failed events
- Filters for amounts > \$5000
- Sends alert to #finance-alerts on Slack

 Risk Level: MEDIUM

 Requires: Finance team approval (company policy)

 Compliance: SOX audit trail enabled

[Test in Sandbox] [Request Approval]"

4. User clicks "Test in Sandbox"

- Simulated charge failure flows through
- User sees Slack message preview
- Evidence trail generated

5. User clicks "Request Approval"

- Finance manager gets notification
- Reviews workflow + risk assessment
- Approves with comment

6. Workflow goes live

- Dashboard shows new agent with risk badge
- Every action logged to Evidence Vault
- User can pull SOX report anytime

What Makes This Different

Feature	Zapier/Make	Our Platform
Connectors	✓ Yes	✓ Yes
Visual builder	✓ Yes	✓ Yes
AI assistance	⚠ Basic	✓ Full orchestration
Risk scoring	✗ No	✓ Per-action
Policy enforcement	✗ No	✓ Real-time
Approval workflows	✗ No	✓ Built-in
Audit trail	⚠ Basic logs	✓ Hash-chained evidence
Compliance reports	✗ No	✓ SOX, HIPAA, FedRAMP
For regulated industries	✗ No	✓ Purpose-built

Update Todos

Building Connector SDK

Build Payment Connectors (Stripe, PayPal, Square, Braintree)

Build CRM Connectors (Salesforce, HubSpot, Zoho)

Build Communication Connectors (Slack, Teams, Email, Twilio)

Build Cloud Connectors (AWS, GCP, Azure)

Build Database Connectors (PostgreSQL, MongoDB, MySQL)

Build Ticketing Connectors (Jira, ServiceNow, Zendesk)

Build Industry Connectors (Healthcare FHIR, Finance Plaid/Yodlee, Energy SCADA)

Build AI Orchestrator (intent parser, workflow designer)

Redesign UI - Light futuristic theme with dark contrasts

Build Governance Center UI